Chapter Notes: Exploring Magnets

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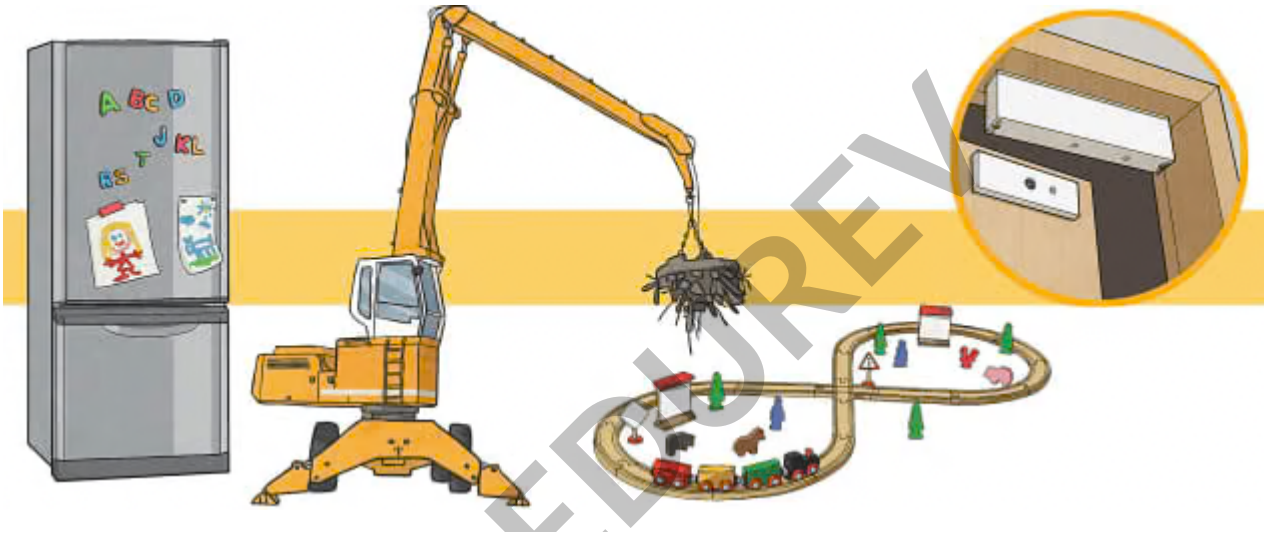
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Introduction

Imagine walking into a scrapyard full of metal junk. You see a large crane moving over a pile, and suddenly pieces of iron start sticking to it as if by magic. This happens because the **crane has a magnet** attached to it.

Magnets are not used only in cranes. They are also used in stickers that stick to refrigerators, magnetic catches that help boxes and cupboards close tightly, toys, speakers and many other objects. Magnets help **hold things together** and move certain materials without touching them directly.



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Magnetic and Non-magnetic Materials

Look at the tools in a geometry box. Some tools are made of metal, while others are made of plastic or other non-metal materials. What makes these materials different from one another?

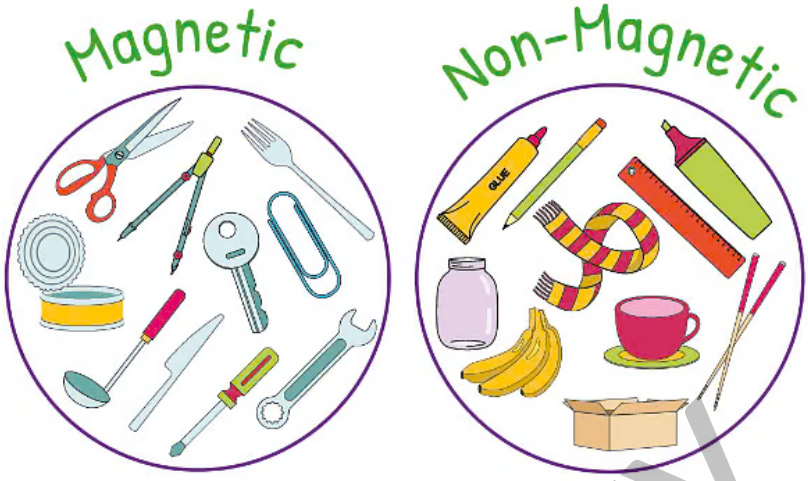


Magnets are useful in many ways in everyday life. Can you name some objects or devices in which magnets are used?

Magnetic and Non-magnetic Materials

Magnetic materials: These are materials that are attracted by a magnet. Iron, nickel and cobalt are common examples. When a magnet is brought near these materials, they may be pulled towards it or stick to it.

Non-magnetic materials: These are materials that are not attracted by a magnet. Wood, plastic, glass and cloth are common examples. They do not stick to a magnet when it is brought near them.



MULTIPLE CHOICE QUESTION

Try yourself: Which of the following materials is considered non-magnetic?

A Iron

B Nickel

C Wood

D Cobalt

View Solution

Poles of a Magnet

Why do you think iron filings stick more to the ends, or poles, of a magnet than to its middle?



When iron filings, which are tiny pieces of iron, are sprinkled around a magnet, they move and arrange themselves along the invisible magnetic field. The magnetic force is strongest near the ends of the magnet. These ends are called the **poles**.

The iron filings show the regions where the **magnetic force** is strongest. Therefore, more filings gather near the poles and fewer filings gather near the middle of a bar magnet.

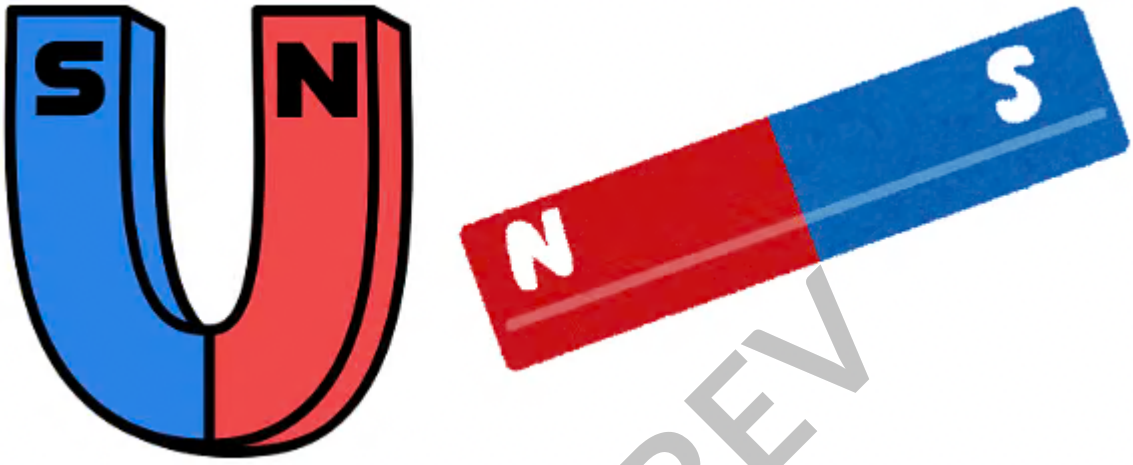
Magnets can have different shapes, such as a straight bar, a horseshoe or a cylinder. In every magnet, the magnetic force is strongest near its poles. Iron filings help us locate these regions.

Why do you think it is impossible to find a magnet with only one pole?



Understanding Magnet Poles

Every magnet has two poles: a North Pole and a South Pole.



The magnetic force is strongest at these poles. If iron filings are sprinkled around a bar magnet, they gather more densely at its ends, showing the locations of the poles.



It is impossible to obtain a magnet with only one pole. Even if a magnet is broken into smaller pieces, every piece will have both a North Pole and a South Pole. A single North Pole or South Pole cannot exist on its own.

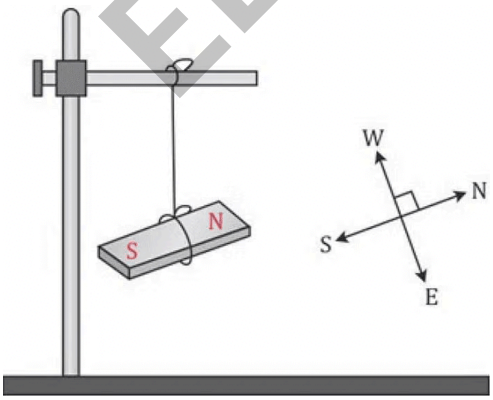
Finding Directions Using a Magnet

Magnets have a special property that makes them useful for finding directions. This property has been known and used by sailors and travellers for many years.

Freely Suspended Magnet

A magnet that is allowed to hang freely settles approximately in the **North-South** direction. The end that points towards the north is called the north-seeking end or **North Pole**. The other end is called the south-seeking end or **South Pole**.

A freely suspended magnet points in the North-South direction after coming to rest. An ordinary iron bar or a wooden stick does not show this behaviour because it is not a magnet.



If you try this with an ordinary iron bar or a wooden scale, it will not behave in the same way because it is not a magnet. Magnets have a North Pole and a South Pole, whatever their shape may be.

In the past, travellers used freely suspended magnets to find their way. Later, people invented the **compass**, which has a magnetised needle that turns to show the North-South direction. The north-seeking end of the needle is often painted a different colour, making it easier to identify the directions.

How It Works?

1. **Suspending the Magnet:** Take a bar magnet and suspend it from a thread. The magnet should hang freely and be able to rotate without any obstruction.



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2. **Alignment:** When the magnet comes to rest, it aligns itself in the North-South direction.

   If the magnet is rotated and released, it again settles in the North-South direction.

Magnetic Compass

A magnetic compass is a device used to find directions. It contains a needle-shaped magnet that can **rotate freely**. When the needle comes to rest, it points in the North-South direction.

How to Use a Magnetic Compass

Place the compass at the location where you want to determine the directions. After a short time, the needle will settle in the North-South direction.

Gently rotate the compass box until the **North and South** markings on the dial match the positions of the needle. The dial will then show the other directions at that location.



Structure of a Magnetic Compass

A magnetic compass is usually a small circular box with a clear cover. Inside the box is a **needle-shaped magnet** placed on a pin fixed at the bottom. The needle is balanced on the pin so that it can rotate freely around it.

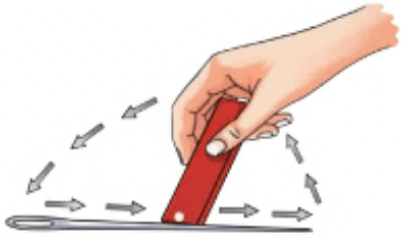
The end of the needle that points towards the north is often painted red. A dial showing the directions is placed below the needle.

Making a Simple Compass

You can make a simple compass at home using a sewing needle, a cork and a bowl of water. Perform this activity with the help of an adult.

1. **Magnetising the Needle:** Stroke a sewing needle several times with a magnet in the same direction to magnetise it.

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2. **Preparing the Cork:** Push the needle through a small piece of cork so that the cork can float on water.
3. **Floating the Needle:** Place the cork in a bowl of water. The needle will rotate and settle in the North-South direction.

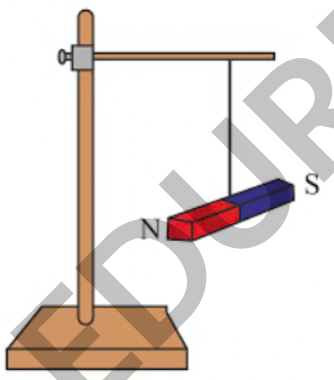


Ancient Indian Navigation Tool: Matsya-Yantra

Long before the modern magnetic compass became widely used, Indians used a similar device for sea navigation. This device was known as the **Matsya-Yantra**, or **Machchh-Yantra**.

It consisted of a fish-shaped piece of magnetised iron placed in a vessel filled with oil. Like a compass needle, it helped people find directions.

Comparison Between a Freely Suspended Magnet and a Magnetic Compass



1. How They Work:

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Freely suspended magnet: When a bar magnet is hung by a thread, it automatically aligns itself along the North-South direction. This happens because the magnet interacts with Earth's magnetic field.

Magnetic compass: A magnetic compass has a small needle that is also a magnet. The needle can rotate freely and points in the North-South direction when it comes to rest, just like a freely suspended magnet.

2. Practical Use:

Freely suspended magnet: This setup is mainly used as a learning tool or in an experiment to demonstrate the directional property of magnets. It is not convenient to carry around for finding directions.

Magnetic compass: A compass is a portable device that can be carried while hiking, travelling or exploring to find directions.

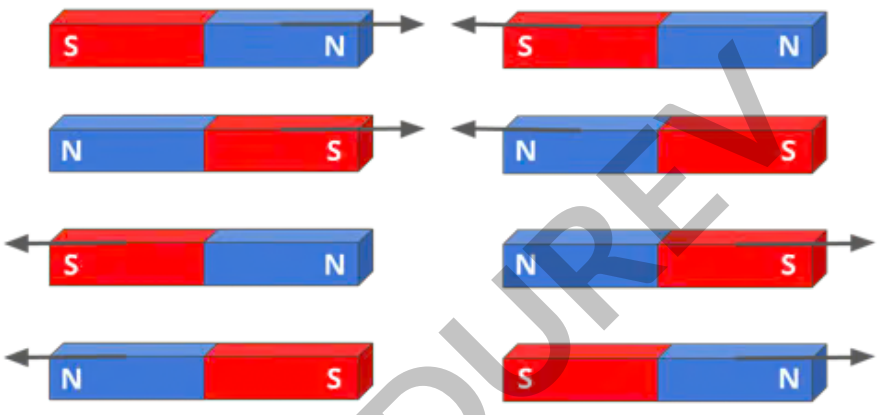
3. Movement:

Freely suspended magnet: It remains still after settling. If it is moved or rotated, it takes some time to settle again in the North-South direction.

Magnetic compass: The needle moves freely and usually settles quickly, making it convenient for finding directions.

Attraction and Repulsion Between Magnets

Magnets can exert forces on one another. When two magnets are brought close together, they either attract or repel each other depending on the arrangement of their poles.



Every magnet has two types of poles: a **North Pole** and a **South Pole**.

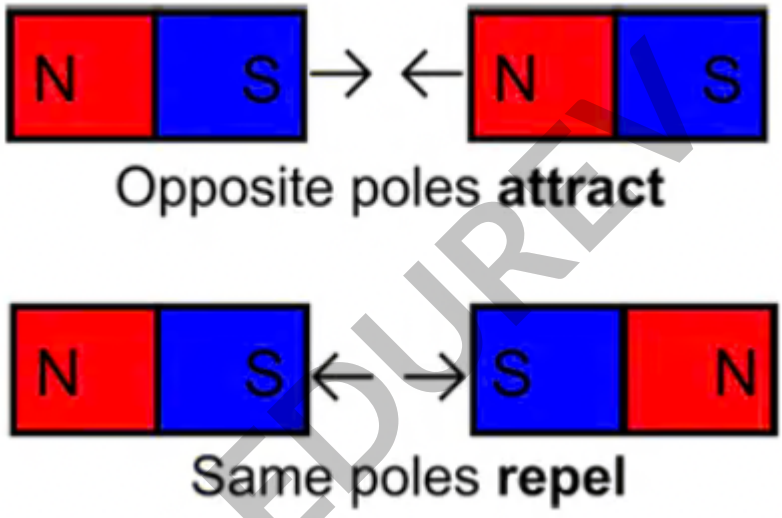
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Opposite poles, such as North and South, attract each other. They pull towards each other.

Similar poles, such as North and North or South and South, repel each other. They push away from each other.

When two magnets with **opposite poles** facing each other are placed close together, they move towards each other because of attraction.

When two magnets with **similar poles** facing each other are placed close together, they move away from each other because of repulsion.



Changing the orientation of the poles, such as positioning the rear of one toy car towards the front of another, changes the direction of movement according to the principles of attraction and repulsion.

This principle can be observed by **suspending a magnet** and studying how it interacts with another magnet. The experiment demonstrates that **opposite poles attract** and **similar poles repel**.

Example: Suppose two toy cars each have a bar magnet. If the South Pole of the magnet on one car faces the North Pole of the magnet on the other car, the cars move towards each other because of attraction. If similar poles face each other, the cars repel and move apart.

MULTIPLE CHOICE QUESTION

Try yourself: Which of the following best describes what happens when opposite poles of two magnets are brought close together?

A They attract each other

B They repel each other

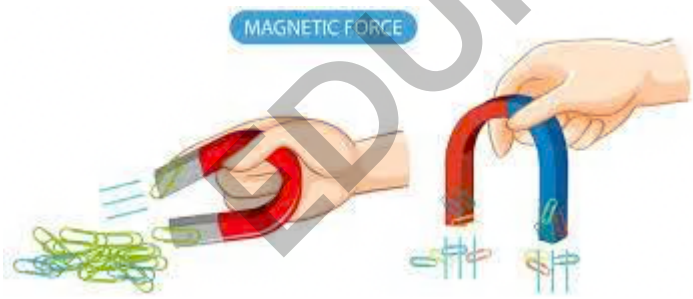
C They do not interact

D They spin uncontrollably

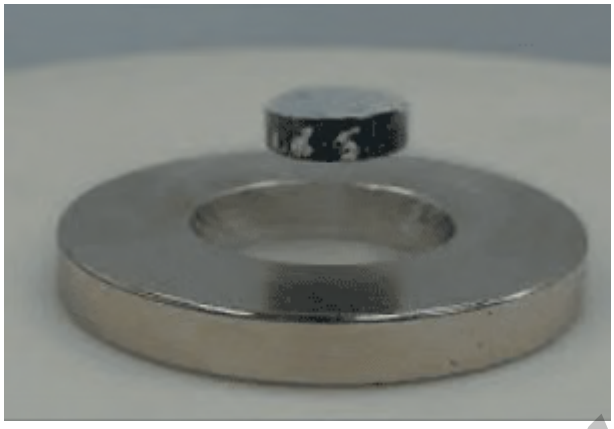
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Fun with Magnets

Can you think of any objects at home that could be used in fun magnet activities?



How could you use magnets to make an object float or hover in the air? Is this possible with the magnets you have?

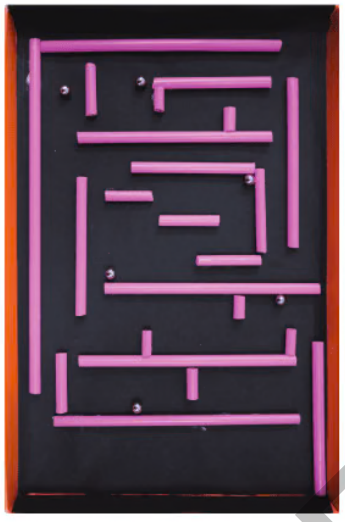


Applications of Magnets in Daily Life

Magnetic garlands: Magnets can be used in decorations to hold or hang objects in a creative display.



Magnetic mazes: In some games, magnets are used to guide steel balls through a maze.



Removing a steel paper clip from water: A steel paper clip can be lifted from water using a magnet without getting your fingers or the magnet wet.



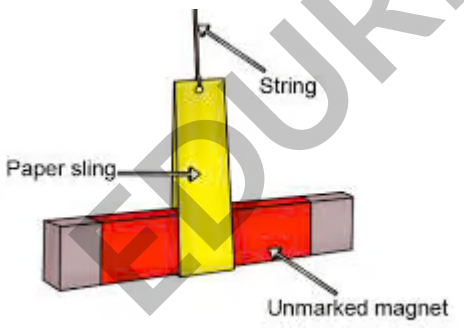
Magnetic effects in experiments: Magnets can demonstrate how magnetic fields influence certain objects without direct contact.





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Identifying the Poles of a Magnet

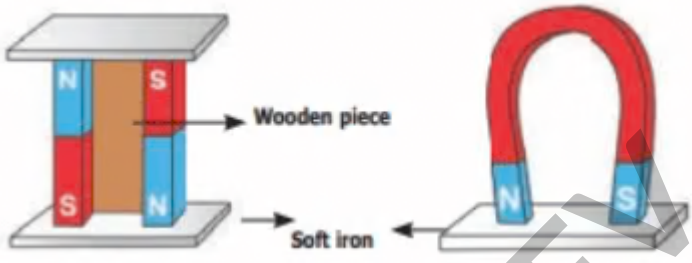


In some magnets, the North and South poles are labelled with the letters **N** and **S**.

In some other magnets, a white dot is used to indicate the North Pole.

Sometimes, the North Pole is painted red and the South Pole is painted blue to make the poles easier to identify.

How to Keep Magnets Safe?



To keep magnets safe, follow these guidelines:

Proper storage: Store bar magnets in pairs with unlike poles on the same side. Place a piece of wood between the magnets and use two pieces of soft iron as keepers across their ends.

Avoid damage: Do not heat, drop or hammer magnets, as these actions can weaken them.

Keep magnets away from electronic devices: Avoid placing strong magnets near mobile phones, remote controls or other electronic devices because magnets may interfere with their operation.

Keywords

Magnetic materials: Substances such as iron, nickel and cobalt that are attracted by magnets.

Non-magnetic materials: Substances such as wood, plastic and glass that are not attracted by magnets.

Poles of a magnet: The two ends of a magnet where the magnetic force is strongest. They are called the North Pole and the South Pole.

Attraction and repulsion: The forces between magnets. Opposite poles attract and similar poles repel.

Compass: A device that uses a magnetised needle to show directions by aligning with Earth's magnetic field.